Line and Scatter Plots

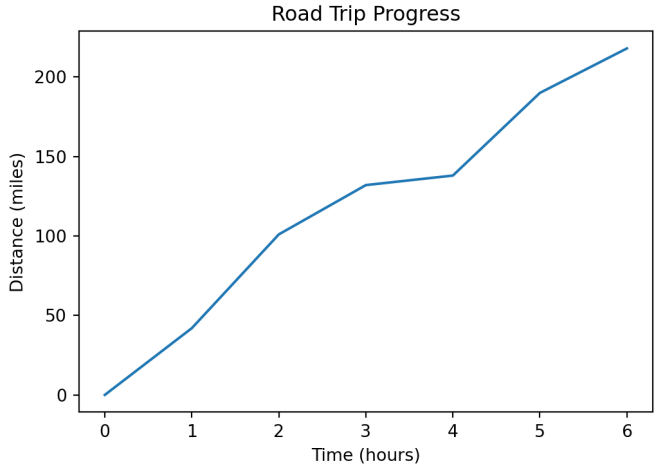
Graphs are fundamental tools for inspecting and communicating with data. Start by importing the `pyplot` plotting library:

```
import matplotlib.pyplot as plt
```

```
t = [ 0,  1,   2,   3,   4,   5,   6]   # Time in hours
d = [ 0, 42, 101, 132, 138, 190, 218]   # Distance in miles

plt.plot(t,d)

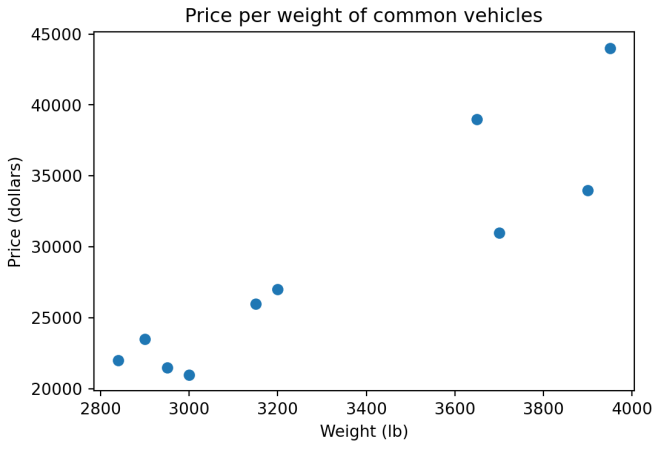
plt.xlabel('Time (hours)')
plt.ylabel('Distance (miles)')
plt.title('Road Trip Progress')
plt.show()
```




```
# Weight in pounds, price in dollars
w = [ 2840, 2900, 3000, 2950, 3150, 3200,
                   3700, 3900, 3650, 3950]
p = [ 22000, 23500, 21000, 21500, 26000,
        27000, 31000, 34000, 39000, 44000]

plt.scatter(w,p)

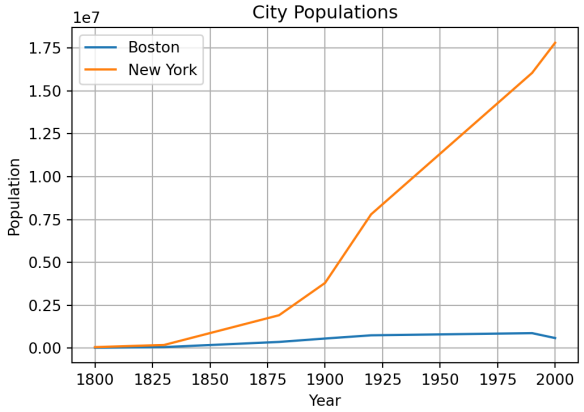
plt.xlabel('Weight (lb)')
plt.ylabel('Price (dollars)')
plt.title('Price per weight of common vehicles')
plt.show()
```




```
# Data
year = [1800, 1830, 1880, 1900, 1920, 1990, 2000]
Boston = [24937, 61392, 362839, 560862, 748060, 874283, 589141]
NewYork = [60000, 185000, 1919000, 3802000,
                7798000, 16044000, 17800000]

# Plot both populations
plt.plot(year, Boston, label='Boston')
plt.plot(year, NewYork, label='New York')

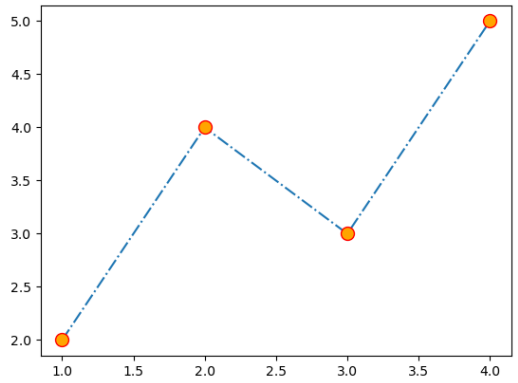
# Display with labels
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('City Populations')
plt.grid()
plt.legend()
plt.show()
```



Plotting with Style

```
x = [1, 2, 3, 4]
y = [2, 4, 3, 5]

plt.plot( x, y,
          linestyle='-.',
          marker='o',
          markeredgecolor='red',
          markerfacecolor='orange',
          markersize=10
        )
plt.show()
```

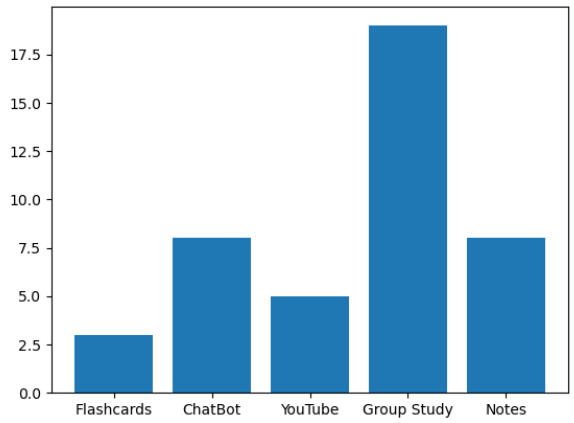


Marker styles: `'.'`, `'*'`, `'+'`, `'x'`, `'o'`, `'s'`
Line styles: `'-'`, `'--'`, `'-.'`, `':'`, `''`

Bar Chart

```
study_methods = ['Flashcards', 'ChatBot', 'YouTube',
                 'Group Study', 'Notes']
counts = [3, 8, 5, 19, 8]

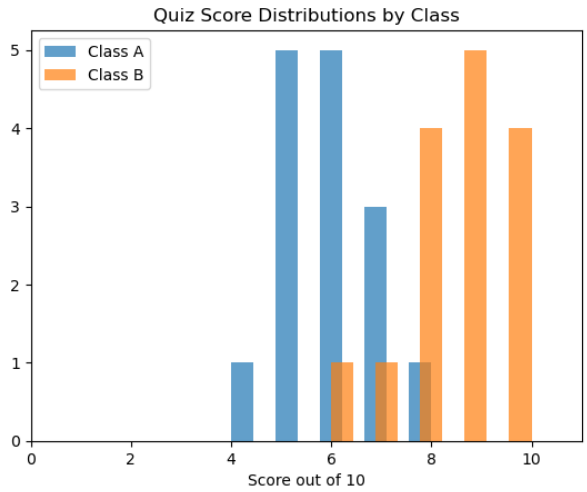
plt.bar(study_methods, counts)
plt.show()
```




```
class_A = [5, 6, 5, 7, 6, 5, 8, 6, 7, 5, 4, 6, 7, 5, 6]
class_B = [9, 8, 10, 9, 10, 8, 9, 10, 6, 8, 9, 7, 8, 10, 9]

plt.hist(class_A, alpha=0.7, bins=9, label='Class A')
plt.hist(class_B, alpha=0.7, bins=9, label='Class B')

plt.legend()
plt.xlabel('Score out of 10')
plt.title('Quiz Score Distributions by Class')
plt.show()
```

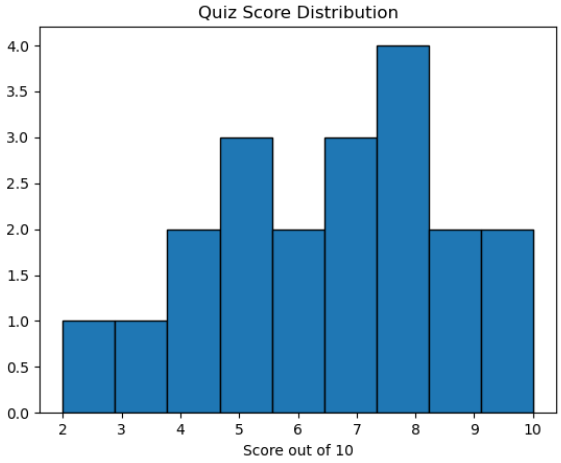


Plotting Distributions

```
scores = [ 2, 3, 4, 4, 5, 5, 5, 6, 6, 7,
           7, 7, 8, 8, 8, 8, 9, 9, 10, 10 ]

plt.hist(scores, bins=9, edgecolor='black')

plt.xlabel('Score out of 10')
plt.title('Quiz Score Distribution')
plt.show()
```

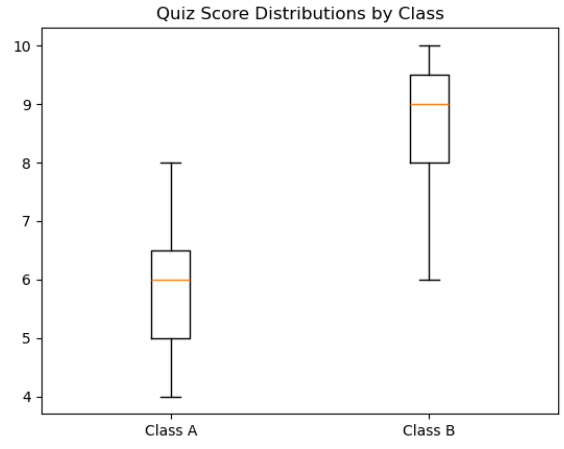


Without specifying, the default number of bins in a histogram is 10 and there is no edge color.

```
class_A = [5, 6, 5, 7, 6, 5, 8, 6, 7, 5, 4, 6, 7, 5, 6]
class_B = [9, 8, 10, 9, 10, 8, 9, 10, 6, 8, 9, 7, 8, 10, 9]

plt.boxplot(
    [class_A, class_B],
    labels=['Class A', 'Class B']
)

plt.title('Quiz Score Distributions by Class')
plt.show()
```



There are many customizations that can be implemented for the plots above. See the matplotlib pyplot documentation.